

Purpose

This document provides a structured set of prompt templates to guide users in generating **synthetic, machine-learning-ready datasets** across three core ML task types:

- Classification
- Clustering
- Association Rule Learning

It uses a **two-level prompt approach**:

- Level 1: Ask for the domain of interest
- Level 2: Automatically suggest dataset parameters (e.g., number of samples, features, types), which the user can confirm to trigger data generation

This approach ensures ease of use for both non-technical users and ML practitioners who need quick, balanced, high-quality datasets for experimentation or demos.

Related Video

Watch this video for a walkthrough or demo:

YouTube Video: Data mining with WEKA complete reference || WEKA crash course

<https://www.youtube.com/watch?v=12eWvSvOqF4&list=PLrP-pAdcAQnux-B5A3zbzhbZ98wNezsjM&index=2>

TWO-LEVEL GUIDED PROMPT FOR SYNTHETIC DATA GENERATION

1. Classification based on supervised learning

LEVEL 1: User Input — Just the Domain

"I want to generate a synthetic dataset for a classification task. The domain I'm interested in is:

[👉 User enters domain: e.g., customer feedback, loan approval, medical diagnosis, fraud detection, etc.]

Please suggest suitable dataset details like class labels, number of samples, features, and feature types for this domain."

LEVEL 2: ChatGPT Responds with Suggestions

"Based on the domain 'XYZ', here are suggested details for your classification dataset:"

Classification goal: [e.g., Predict if a loan should be approved or not]

Number of classes: 2 (e.g., Approved, Rejected)

Number of samples: 1000 (balanced across classes)

Number of features: 8

Feature types: Mixed (numerical + categorical)

Feature suggestions:

'age': Age of applicant

'income': Monthly income

'employment_status': (e.g., employed, unemployed, self-employed)

'credit_score': Credit score on a scale

'loan_amount': Requested loan amount

'existing_loans_count': Number of existing loans

'residence_type': Own, rent, with family

'has_criminal_record': Binary flag (0 or 1)

Target column: 'loan_status' with values: 'Approved', 'Rejected'

Balance: Dataset will be evenly distributed across classes.

Data quality: Designed to avoid overfitting or underfitting, and suitable for training ML models.

Do you want to proceed and generate this dataset in CSV format?

If the User Says "Yes"

ChatGPT proceeds to generate the dataset as a CSV table.

2. Clustering using unsupervised learning

LEVEL 1: User Input — Just the Domain

"I want to generate a synthetic dataset for an unsupervised machine learning task (clustering). The domain I'm interested in is:

[👉 User enters domain: e.g., customer segmentation, health patterns, product usage behavior, etc.]

Please suggest suitable dataset structure, number of clusters, number of features, and feature types for this domain."

LEVEL 2: ChatGPT Responds with Suggestions

"Based on the domain 'XYZ', here are the suggested details for your clustering dataset:"

Clustering goal: [e.g., Group customers by purchasing behavior]

Estimated number of clusters: 3 (can be adjusted)

Number of samples: 1000

Number of features: 6

Feature types: Numerical / Categorical / Mixed

Feature suggestions:

'age': Age of individual

'monthly_spending': Average monthly spend

'region': Region or location category

'visits_per_month': Number of visits per month

'preferred_channel': Online, In-store, or Both

'loyalty_score': Score from 0 to 1

Target: No explicit target; goal is to explore and identify natural groupings

Data quality: Well-separated clusters for clear results but still realistic and varied

Would you like me to generate this synthetic dataset in CSV format?

If the User Says "Yes"

ChatGPT generates the clustering dataset as a CSV table — no label column, but data structured to support useful unsupervised learning.

3. Association rule learning

LEVEL 1: User Input — Just the Domain

"I want to generate a synthetic dataset for association rule mining. The domain I'm interested in is:

[👉 User enters domain: e.g., retail transactions, medical symptoms, online behavior, etc.]

Please suggest suitable dataset structure, item types, and transaction formats for this domain."

LEVEL 2: ChatGPT Responds with Suggestions

"Based on the domain 'XYZ', here are suggested details for your association rule mining dataset:"

Use case: [e.g., Identify which products are commonly bought together]

Number of transactions: 1000

Average items per transaction: 5

Item pool (frequent items):

[e.g., for grocery store: bread, milk, eggs, cheese, cereal, soda, fruit, etc.]

Format:

Each transaction is a list of items (CSV row format)

Example row: bread, milk, cereal, fruit

Support for patterns: Some items co-occur more frequently to support realistic rule mining

Noise: Light randomness to avoid trivial patterns

Would you like me to generate this synthetic transaction dataset for association rule mining in CSV format?

If the User Says "Yes"

ChatGPT generates a dataset.